

PHYLOGEOGRAPHY WORKSHOP

INTRODUCTION

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Phylogeography Workshop

Day 1

1 WHAT IS PHYLOGEOGRAPHY?

2 OVERVIEW OF THIS WORKSHOP

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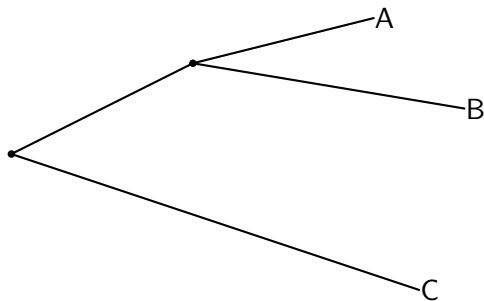
FIRST, PHYLOGENIES:

Phylogenies describe ancestral relationships among organisms

Our samples could be representatives of species (macroevolution),
 or they could represent sequenced viruses from infections
 (microevolution)

HOW DO WE INTERPRET TREES?

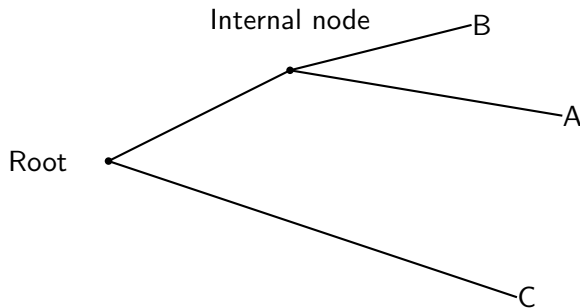
Trees have a **topology** tells us about the ancestry relationships of the taxa (species, infections, etc.)



This tree's topology can be written: ((A,B),C)

HOW DO WE INTERPRET TREES?

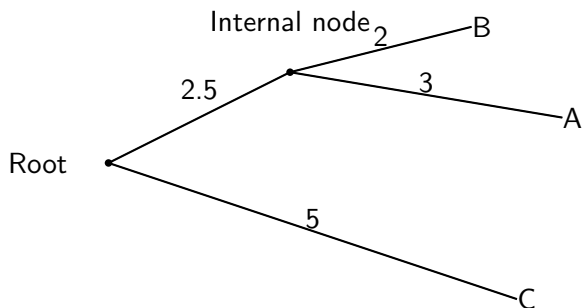
Trees have a **topology** tells us about the ancestry relationships of the taxa indicated at the leaves (species, infections, etc.)



This tree's topology also can be written: $((A,B),C)$

HOW DO WE INTERPRET TREES?

The branches/edges of trees have lengths. Units of branch length are often substitutions/site/year, but molecular clock models can rescale these to units of time



These branch lengths can be used to define **patristic (or phylogenetic) distances** between the taxa. For example:

$$d(A, B) = 2 + 3,$$

$$d(\text{Internal node}, C) = 2.5 + 5$$

LOTS OF THINGS PHYLOGENIES CAN BE USED FOR

In addition to describing evolutionary relationships among organisms, phylogenies are useful for understanding evolution of traits

- phylogenetic comparative methods
- discrete traits

Phylogeography can be treated in discrete or continuous space, but we will focus on discrete-state phylogeography in this workshop

HUMAN CULTURAL EVOLUTION

Phylogeographic analysis of humans supports movement of populations out of Africa to the rest of the world¹

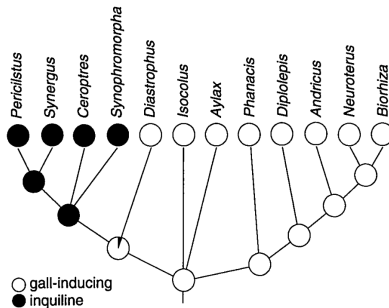
Useful for exploring hypotheses related to cultural evolution

- associations of language with phylogeny
- hunter-gatherer societies: more matriarchal, similar mitochondrial DNA ~ similar language
- agricultural societies: more patriarchal, Y chromosome similarity ~ similar language

¹Alexander H Harcourt. "Human phylogeography and diversity". In: *Proceedings of the National Academy of Sciences* 113.29 (2016), pp. 8072–8078.

How did inquilinism in wasps evolve?²

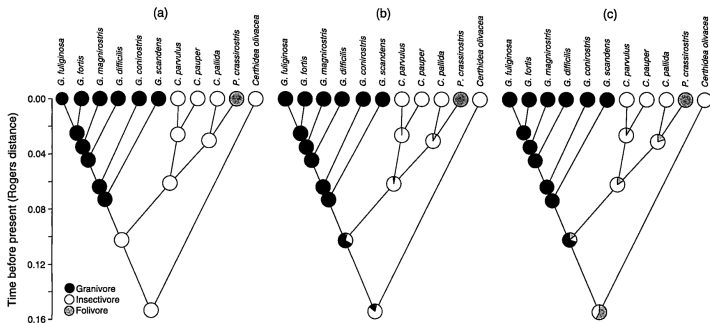
Some wasps induce galls in plants to lay their eggs; **iniquilism** is when some species of wasps steal galls from other wasps



²Dolph Schluter et al. "Likelihood of ancestor states in adaptive radiation". In: *Evolution* 51.6 (1997), pp. 1699–1711.

EVOLUTION OF DIET IN BIRDS

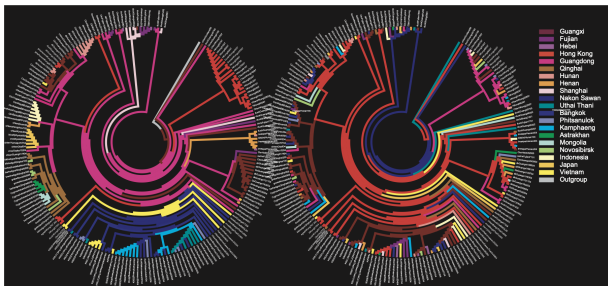
How did the modern diets of finches evolve?³



³Dolph Schluter et al. "Likelihood of ancestor states in adaptive radiation".

In: *Evolution* 51.6 (1997), pp. 1699–1711.

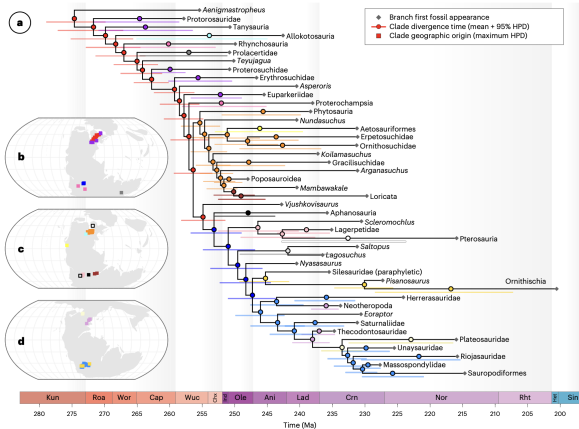
How does Influenza H5N1 move around the world?⁴



⁴Robert G Wallace et al. "A statistical phylogeography of influenza A H5N1". In: *Proceedings of the National Academy of Sciences* 104.11 (2007), pp. 4473–4478.

REPTILES

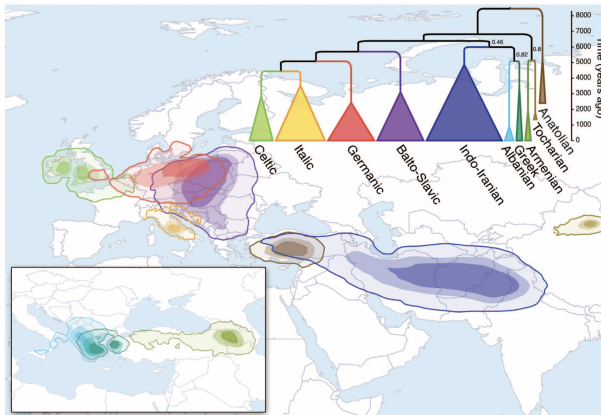
Where do particular species originate?⁵



⁵ Joseph T Flannery-Sutherland et al. "Landscape-explicit phylogeography illuminates the ecographic radiation of early archosauromorph reptiles". In: *Nature Ecology & Evolution* (2025), pp. 1–15.

LANGUAGE

How do languages evolve?⁶



⁶Remco Bouckaert et al. "Mapping the origins and expansion of the Indo-European language family". In: *Science* 337.6097 (2012), pp. 957–960. 

Synonyms for phylogeography (as far as this workshop is concerned):

- ancestral character estimation
- ancestral state reconstruction
- ancestral trait reconstruction

Phylogeography is a bit of a misnomer: really, the methods we will be learning about can be used to study evolution of any discrete trait

There are methods for analyzing continuous traits, but we will not explore those in this workshop

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PLAN FOR THE WEEK

We will cover a lot of topics, but the materials on these web pages are designed to help you make the most of this week. The days are organized roughly as follows:

- Day 1: Markov models, ancestral character estimation, and stochastic character mapping, and basic algorithms in ancestral state inference
- Day 2: Introduction to stochastic processes in phylodynamics: the Yule process, the birth-death process, and versions of these models with incomplete sampling
- Day 3: State-dependent speciation and extinction (SSE) models (such as BiSSE, MuSSE), and Sampling-Aware Ancestral State Inference (SAASI)
- Day 4: Extending these methods to a Bayesian setting
- Day 5: Bayesian Stochastic Search Variable Selection and comparing methods against each other

SOME COMMENTS ABOUT HOW THIS WORKSHOP WILL FUNCTION

As you may notice from the website, modules for this workshop contain either slides or code demos.

Feel free to come and go as necessary to meet your obligations for classes, etc.

The practical sessions in R are designed to help you gain familiarity with the different methods we are learning about in this course

The codes are designed to work with simulated trees, but you may wish to edit things to load example phylogenies from real datasets

As we progress through the week, there will be opportunities to compare results from the different types of analyses available to us

NEXT:

Today, we will be working in R. You will need the following packages installed:

- ape
- expm
- phytools
- coda